TigerMed

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AWS migration proposal for a fictional healthcare platform operating across three regions

Project brief

The assignment: propose an AWS migration for a fictional healthcare platform using Windows, IIS and SQL Server, with regional data controls and reliable document processing.

01

Regional

patient-data ownership

02

Separate

document processing

03

Recovery

without a second active estate

Baseline infrastructure

02

Web servers

2 vCPU · 4 GB each

02

Application servers

4 vCPU · 16 GB each

01

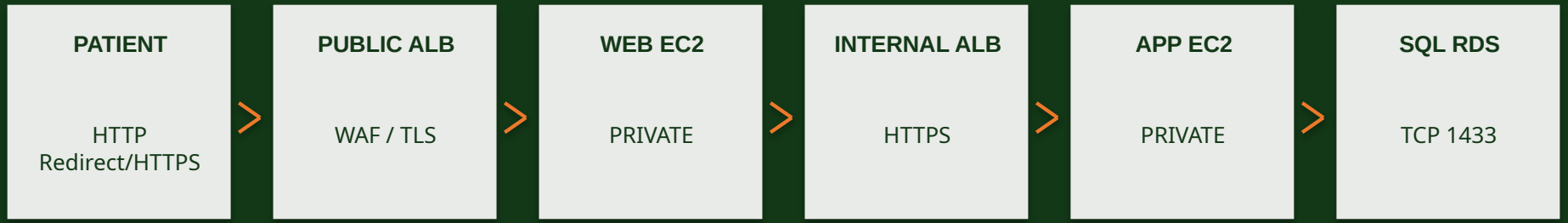
SQL Server

8 vCPU · 32 GB · 5 TB

Assumed physical Windows/IIS servers and SQL Server Standard; demand and growth require validation.

Routing

Only the public load balancer accepts inbound internet traffic; web, app and data tiers remain private.



When dealing with documents:

S3 upload → SQS queue → private processing workers → protected S3 results (KMS Encrypted)

Regional footprint: production and recovery

US

N. Virginia

Production · us-east-1

10.10.0.0/16

RECOVERY

Ohio · us-east-2

10.60.0.0/16

EUROPE

Frankfurt

Production · eu-central-1

10.50.0.0/16

RECOVERY

Zurich · eu-central-2

10.70.0.0/16

APAC

Singapore

Production · ap-southeast-1

10.40.0.0/16

RECOVERY

Sydney · ap-southeast-2

10.80.0.0/16

Additional US VPCs: Testing 10.20.0.0/16 and Development 10.30.0.0/16 which are both in us-east-1.

Each patient has their own home region

Regional routing improves access and clinical records stay where TigerMed is authorized to store them

01

Assign

Home Region at onboarding

02

Enforce

Authorization in the application

03

Restrict

Cross-Region data transfers

Note: Geographic DNS alone does not determine legal data residency and additional review may be necessary.

Production VPC in two AZs

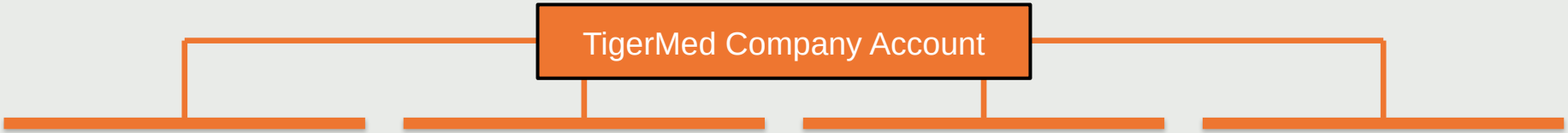
Ten subnets per Region: five isolated subnet types repeated across AZ A and AZ B.



Internet ALB + NAT	2 subnets
Windows IIS + EC2	2 subnets
Internal ALB + EC2	2 subnets
SQS workers on EC2	2 subnets
SQL Server RDS	2 subnets

No direct inbound internet routes to private subnets. Recovery VPCs omit dedicated processing subnets.

Identity and Access



SYSTEMS ADMIN

Infrastructure administrators

Manage EC2, VPC, load balancers and scaling

DATABASE ADMIN

Database administrators

Manage RDS and approved database operations

MONITORING

Monitoring team

Read-only observability with no edits or deletion permissions

APP ROLE (S3)

Application service role

Read and write TigerMed S3 buckets only

Workforce authentication

PASSWORDS	Strong passphrases with controls against compromised passwords
ROTATION	Rotate credentials after suspected compromise or policy requirement
CONSOLE	MFA and centralized workforce access through IAM Identity Center
CLI / API	Prefer temporary role credentials over permanent administrator keys

AWS workforce access does not replace patient/doctor sign-in or record-level authorization.

Network Architecture

COMPONENT	INBOUND	ALLOWED FROM
Public ALB	HTTPS 443, HTTP 80 redirect -> HTTPS	Internet
Web EC2	HTTPS 443	Public ALB security group
Internal ALB	HTTPS 443	Web EC2 security group
Application EC2	HTTPS 443	Internal ALB security group
RDS SQL Server	TCP 1433	Application EC2 security group
Processing EC2	No inbound	Outbound S3 / SQS endpoints

No public IPs for EC2, workers or RDS · Systems Manager for administration · AWS WAF on public ALBs.

Proposed instance sizing

TIER	AWS RESOURCE	PER-REGION START	ROLE
Web	EC2 t3.medium	2 × 2 vCPU / 4 GiB	Windows + IIS
Application	EC2 m6i.xlarge	2 × 4 vCPU / 16 GiB	Windows + IIS
Database	RDS db.m6i.2xlarge	1 primary + 1 standby	SQL Server SE
Document jobs	EC2 c6i.xlarge	0–4 on demand	Worker fleet

Illustrative sizing; validate through load tests. S3 stores files, EBS instance volumes and RDS clinical data.

Security Groups

Group Name	Type	Port	Source	Description
Public ALB	HTTPS HTTP	443 80	Internet (0.0.0.0/0)	Secure Public Entry Point, HTTP redirects to HTTPS
Web EC2	HTTPS	443	Public ALB SG	Only receives traffic from public ALB
Internal ALB	HTTPS	443	Web EC2 SG	Routes web requests to app tier
Application EC2	HTTPS	443	Internal ALB SG	Private Application Access
Database RDS	TCP	1433	Application EC2 SG	Can only access from Application Tier
Document workers	None		No Inbound	Access S3/SQS via endpoints

Backup and Disaster Recovery

US PRODUCTION

Ohio recovery

Virginia → Ohio

EU PRODUCTION

Zurich recovery

Frankfurt → Zurich

APAC PRODUCTION

Sydney recovery

Singapore → Sydney

Recovery Regions are proposed backup/restore targets; regional failover requires a tested runbook.

BACKUP PLAN

SERVERS

Recreated with launch templates and Auto Scaling

DATABASE

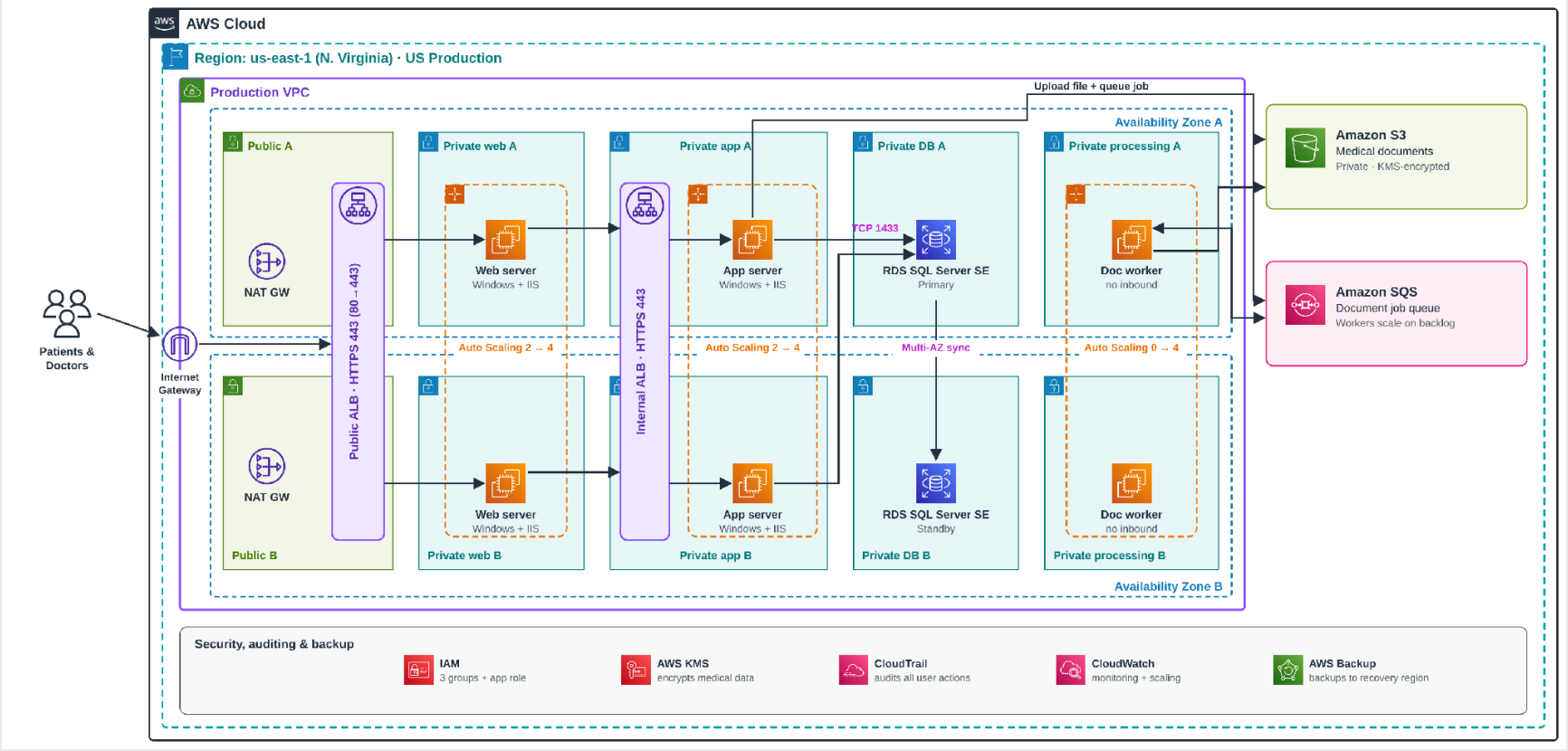
RDS backups copied to recovery region

MEDICAL FILES

Protected S3 copies in recovery region

Reference architecture

Illustrative US production design; equivalent regional patterns proposed for Frankfurt and Singapore.



High Availability and Auto Scaling

WEB

2 → 4

per Region · doubles today

ALB requests / target

APPLICATION

2 → 4

per Region · doubles today

CPU utilization

DOCUMENT

0 → 4

per Region

SQS queue backlog

DATABASE

1 + 1

primary + standby

Multi-AZ auto failover

Self-healing: EC2 status checks catch OS failures and ELB health checks catch application failures; Auto Scaling replaces the server automatically from a launch template whose User Data installs IIS.

Privacy and Encryption

PRIVATE BY DEFAULT

EC2 and RDS stay off the public internet.

ENCRYPTED

In transit: TLS
At rest: AWS KMS for S3, EBS and RDS

FILE VALIDATION

Uploaded documents are validated before clinical release.

PATIENT-SPECIFIC

Application checks each doctor/patient record authorization.

Auditing

INFRASTRUCTURE TRAIL

- CloudTrail: console, CLI, SDK and API calls
- Config: resource changes
- VPC Flow Logs and GuardDuty findings
- S3 data events for sensitive document access

CLINICAL ACCESS LEDGER

- Who viewed or edited a patient record
- When and from which assigned Region
- Authorization result and action type
- Protected retention without medical content

Logs encrypted and retained in controlled S3 storage

Illustrative monthly budget

~$23k

per month

Planning estimate for the three production regions, recovery copies, and dev/test environments.

This is a classroom estimate, not a validated AWS quote. Actual licensing, usage, transfer and recovery costs require pricing analysis.

Web servers (6)	$350
App servers (6)	$1,950
Database: RDS Multi-AZ (3)	$16,600
Document workers	$1,000
Load balancers (6)	$150
NAT gateways (6)	$250
S3 + SQS	$200
Monitoring + security	$150
Backups + recovery copies	$1,000
Dev + Test environments	$1,200
Total	**$22,850**

Architecture tradeoffs

A lower-cost first phase changes which requirements can be met at launch.

	Original proposal	Smaller initial rollout
COVERAGE	US, Europe and Asia-Pacific live	US first; other regions deferred
RECOVERY	Separate recovery regions and copies	Multi-AZ locally plus backup restoration
COST PROFILE	Multiple regional stacks and DB licenses	Fewer active resources at launch

Tradeoff: A US-first launch is only viable if EU/APAC access and data-residency requirements are not needed on day one.

Questions?

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